

TARAN WARREN

Machine Learning & AI Engineer

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PROFILE

First-Class AI & Robotics graduate specialising in deep learning, computer vision, and end-to-end automated systems. I focus on getting models and pipelines to work reliably in the real world — handling noisy data, exposing evaluation pitfalls, and building systems that run unattended. Seeking a Machine Learning / AI Engineer role taking projects from research through to production.

EDUCATION

BSc (Hons) Artificial Intelligence & Robotics — University of Staffordshire 2023 – 2026

First-Class Honours

- Dissertation graded **First Class**: a multi-label deep learning pipeline (EfficientNet-B1) for classifying six intracranial haemorrhage types from CT scans, including identifying and correcting a slice-leakage evaluation flaw that had inflated reported accuracy.

Level 3 Diploma, Computer Games & Programming — Truro and Penwith College 2020 – 2022

Grade: Distinction

- Foundations in programming, 3D modelling and animation; active in the college Esports society.

SELECTED PROJECTS

Intracranial Haemorrhage Detection (Dissertation) 2026

Python · PyTorch · EfficientNet-B1 · Computer Vision

- Built a full medical-imaging pipeline — DICOM preprocessing, CT windowing, RGB channel engineering, normalisation and tensor caching — cutting per-epoch training time from ~20 minutes to under 5.
- Handled severe class imbalance (rarest class 0.5% of data) with class weighting, stratified sampling and rare-class oversampling; used transfer learning with two-stage fine-tuning, AMP and CUDA prefetching.
- Identified **slice leakage** as the cause of an artificially high 0.96 macro-F1; a corrected study-level split gave an honest 0.856 macro-AUC — demonstrating that evaluation methodology matters as much as model architecture.

Automated YouTube Content Pipeline (“Towards Terra”) 2025 – Present

Python · LLM APIs · Stable Diffusion / ComfyUI · TTS · ffmpeg · YouTube Data API

- Designed and built a fully automated pipeline that writes, illustrates, narrates, assembles, captions and uploads cinematic sci-fi videos on a daily schedule with almost no manual input.
- Integrated LLM story generation (with a batch-API cost optimisation), local Stable Diffusion image generation, multi-voice text-to-speech, local Whisper captioning and scheduled uploads with automated AI-content disclosure.
- Engineered fallbacks for every external dependency and a vision-model QC step to catch image drift; the channel has surpassed **10,000 views** of automatically produced content.

Animatronic Pterodactyl (Group Robotics Project) 2024

Blender · 3D Printing (PETG) · Mechanical Design

- Took an articulating animatronic prototype from 2D concept through virtual prototyping to a working 3D-printed model, with wings actuated by a ligament-inspired internal pulley system and print-in-place joints to cut assembly complexity.

EXPERIENCE

Quality Assurance & Bottling — Artisan Vinegar Jul 2022 – Sep 2024

Part-time · Coverack, Cornwall

- Improved and developed procedures to increase the efficiency and quality of the product.

TECHNICAL SKILLS

Languages & ML: Python, PyTorch, deep learning, CNNs (EfficientNet/ResNet), computer vision, transfer learning, model evaluation (ROC-AUC, F1, calibration).

Tools & Systems: Stable Diffusion / ComfyUI, LLM & cloud APIs, ffmpeg, Whisper, Git, IntelliJ IDEA, Replit, Blender, 3D printing.

Strengths: End-to-end automation, pipeline reliability, data preprocessing, debugging subtle failure modes, self-directed delivery.